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CLINICAL MANIFESTATIONS

POSTER PRESENTATION

# BDNF-mediated improvements in cognition after computerized cognitive training

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## Abstract

**Background:** Evidence suggests that engagement in cognitively stimulating activities may reduce risk of cognitive deterioration and dementia. Mechanisms underlying these observations remain to be determined. Animal research indicates that brain-derived neurotrophic factor (BDNF) plays an important role in neural plasticity, neurogenesis, synaptic growth, and cognitive performance. Working under the assumption that augmented BDNF levels benefit brain function, investigations of human subjects have focused on determining whether physical or mental activities can increase BDNF levels. Surprisingly little research has been devoted to linking alterations in BDNF to changes in cognition. This study aimed to determine whether changes in BDNF contribute to the enhancement of cognitive performance after computerized cognitive training (CCT). We have demonstrated that five weeks of CCT emphasizing working memory (WM) was associated with improvement on an untrained test of WM/processing speed, the Digit Symbol Test. Here, we investigated whether this improvement was mediated by changes in BDNF levels.

**Method:** Seventy-five older adults, ages 65-86, at two sites (USA and Sweden) were randomized to either Cogmed Adaptive (n=37) or Non-Adaptive (active control) (n=38) CCT. Under the adaptive CCT, task difficulty was revised on a trial-by-trial basis to create a consistently high level of challenge. Under the active control condition, task load remained at a constant, relatively low level. CCT was performed in the homes of participants, five days per week, over five weeks. Serum BDNF levels and performance on Digit Symbol were measured pre- and post-intervention.

**Result:** Analysis using linear mixed models demonstrated an interaction between time and intervention group for Digit Symbol score ($p$=.036) and BDNF level ($p$=.013) Only the adaptive CCT group demonstrated an augmentation of BDNF level ($p$=.014) and improvement in Digit Symbol scores ($p$<.001) after training. Mediation analysis

  

revealed that changes in BDNF level accounted for the intervention-related improvement in performance on Digit Symbol. Regression analysis indicated that changes in BDNF level predicted changes in Digit Symbol score (p=.010).
**Conclusion:** The results of this investigation suggest that increases in BDNF linked to cognitive training mediate improvement in cognitive performance. Additional research is needed to determine if these findings are specific to WM or are observed in other cognitive domains.